IOT6000U Assessment Brief – 002 Individual Coursework

Module Code	**IOT6000U**
Module Title	Digital Innovation and Product Development
Module Organiser(s)	Jonathan Jackson & Alim Gias
Assessment	002 Individual Coursework
Weighting (%)	80%
Deadline	Tuesday 12th January 2027, 23:59
Duration / Length	3000 words

Assessment Outline

For this assessment, you will showcase the work completed for the area of investigation identified as part of your 001 submission. Make sure to utilise the feedback received from your 001 presentation to enhance your submission for this assessment.

You are required to submit a report of 3000 words (excluding executive summary, reference list, and appendices) and +/- 10% of this word count should enable you to produce work that is appropriately detailed.

Your report should be written and formatted in a manner appropriate to a professional context. Include page numbers, utilise appropriate heading styles, and an automatically generated table of contents or means of navigation.

Pseudonymise, anonymise, or redact any personally identifiable information (PII) included in your evidence throughout your work, including any appendices.

Below is a suggested structure for your report, and word count estimates have been provided to help you ascertain the proportionate level of depth required for each section.

Some specific KSBs have been included in brackets to highlight the relevance of particular assessment elements to your apprenticeship standard:

1. **Cover Sheet** – include Student Name, Student ID, Module and Assessment details, and word count
2. **Executive Summary** – a succinct overview of the whole project and outcomes
3. **Table of Contents**

4. **Introduction and Background (500 words)** – outline the selected area of investigation, summarise related evidence and resources, demonstrate the real-world significance and make the aim of the project clear.
5. **Proposed Solution (500 words)** – formalise details of the solution originally proposed in your 001 presentation **(S3)**.
    - Include any design documentation of relevance to your project. For example: functional and non-functional requirements; user stories; storyboards; prototype visuals; wireframes; data models; architecture diagrams; data flow diagrams; etc…
    - An iterative approach is encouraged so demonstrate where your design documentation has been refined based on new information or feedback over the duration of the project. You can evidence this across any relevant sections of your submission and/or use appendices to include additional evidence.
6. **Proof of Concept (750 words)** – evidence the tangible outputs of the project which may involve code, no/low-code, or a combination of both. The proof of concept (PoC) does not need to be a complete solution. Rather, focus on the more novel aspects of the proposed solution to demonstrate its feasibility **(S4)**.
    - Showcase the most compelling aspect(s) of your approach to developing the PoC, explaining any technical decisions you made along with a rationale (e.g., why you chose a particular AI tool, code library, no-code platform, or cloud provider)
    - Evidence your PoC work through selective screenshots of code, configuration, or user interfaces, as applicable. Utilise appendices for additional evidence to avoid overwhelming the body of your report with too many unexplained visuals.
    - In addition, link to any external sources where they cannot be effectively embedded in your report (e.g., video demo, code repository, online python notebook)
    - As well as evidencing your outputs, evidence your process by showing how you utilised tools such as version control, task trackers, or Generative AI (GenAI) to manage, track, enhance, or document your work **(S6).**
7. **Privacy and Security Evaluation (750 words)** – explain and evidence three specific examples of how your solution addresses data privacy and/or security risks **(B3).** For each example:
    - Identify the potential risk or threat, explain why it is relevant to your solution, and evaluate its likelihood and potential impact.
    - Describe how your solution mitigates against the risk and justify the approach taken.
    - Include appropriate evidence of the mitigations. For example, if you have written code which you say is protected against SQL injection, show the code, and show a screenshot of an HTTP request trying (and failing) to trigger a SQL injection attack.
8. **Conclusions and future plans (500 words)** – reflect on the project as a whole, future opportunities, and your personal learnings.
    - Evaluate the success of the project as a proof of concept, linking your justification back to the original project aim, and propose the most effective

next steps if the project was to be taken forward beyond the scope of this module.

- Critical overview of how feedforward received in 001 was used to review the idea/concept and how this contributed to the overall effectiveness of the final concept.
- Include your reflections on your most significant learnings from engaging with this assessment.
- Explain what KSBs from your apprenticeship standard you feel it has helped you engage with most.

9. **References** – Harvard or IEEE format preferred.
10. **Appendices** – number and name these with sub-headings appropriately e.g. "Appendix 1 – architecture diagram iterations". All appendices should be referred to in the main body of the report to indicate the specific relevance of each appendix.

It is expected that the outputs or artefacts you produce will vary greatly depending on your area of investigation and your subject-specialism, and this flexibility has been built into the assessment criteria for this assessment.

See the "Assessment Criteria" at the end of this document for more detail on how your work will be assessed.

Submission Requirements

Your report submission ***must*** be a text-based report format (.docx or .pdf), not a zip file, and not password protected. Any submissions which circumvent plagiarism checks will be marked as a fail (0).

Your submission must be as self-contained as possible and avoid relying on external links except in cases such as supporting video, live prototypes, or code repositories.

When linking to external resources, ensure they are publicly available. Resources that cannot be accessed will be ignored.

AI Literacy and Generative AI

GREEN 🟢 (AI required)

Use of some AI technology is **required** for at least part of this assessment which has been designed to support you in developing and evidencing AI literacy within the context of your studies.

The following are ***additional*** examples of potential uses of Generative AI (GenAI) that would be appropriate for this assessment, but are not required:

✔ **Creating diagrams, visuals or infographics**	Using AI to generate or refine visual representations based on your own content, analysis, or interpretation. **Your responsibility**: You remain responsible for the accuracy, meaning, and interpretation of any visual material submitted and should not present AI-generated data, results, or analysis as your own.

✔ **Checking and/or improving spelling, grammar and punctuation**	Using AI as a proofreading or editing aid to identify spelling, grammar, punctuation, or typographical errors. **Your responsibility**: You remain responsible for ensuring that any suggested changes accurately reflect your intended meaning and do not introduce new content or ideas.
✔ **Coding support** **(disclosure required)**	Using AI to debug code, explain unfamiliar code, generate code comments, create boilerplate code, or translate code between programming languages. **Your responsibility**: You remain responsible for understanding, evaluating, and appropriately applying any code used. You should be able to explain any algorithmic approaches, implementation decisions and justifications. **You must disclose and evidence which AI coding tool(s) you used, as outlined in the assessment brief above.**
✔ **Agentic AI coding** **(disclosure required)**	Using Agentic AI to develop end-to-end software functionality. **Your responsibility**: You remain responsible for understanding and evaluating the quality (e.g. maintainability, security) and functionality of the code produced. **You must disclose and evidence which agentic AI coding tool(s) you used, as outlined in the assessment brief above.**
✔ **Improving clarity and readability**	Using AI to suggest clearer phrasing, improve expression, or enhance the readability of text. **Your responsibility**: You remain responsible for the meaning, argument, and interpretation presented in your work.

If using spelling or grammar assistive tools, be cautious if using any GenAI features which rewrite non-trivial portions of your work in case your original meaning is lost or your writing ends up being incoherent. You remain fully responsible for the work you submit.

Avoid relying on GenAI to format your references as these tools can often produce references which do not exist which will significantly undermine the credibility of your work and may be considered academic misconduct. Dedicated reference management tools such as EndNote, Zotero or Mendeley can provide more reliable support for this task.

If you are in doubt about the appropriateness of using any specific assistive technology, please discuss with your module organiser.

For more guidance, please see https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/.

Late Submissions and Extenuating Circumstances

Assessments submitted after the published deadline will be marked as late and will receive a penalty, unless you have approved Extenuating Circumstances (EC).

You may submit up to seven calendar days after the deadline but a penalty of 5% of the total marks available will be applied to the assessment for each 24-hour period (or part of it) that the work is late (e.g. five marks deducted each day from a total of 100). After seven calendar days the assessment will be recorded as a non-submission and given a mark of zero.

For more information, you can find the latest Assessment Handbook and Extenuating Circumstances policy via the QMUL policies page here: https://www.qmul.ac.uk/governance-and-legal-services/policy/policies-by-category/

Knowledge, Skills, and Behaviours (Apprenticeship v1.2)

This assessment is designed to help you engage with the following Knowledge, Skills, and Behaviours (KSBs) from your apprenticeship standard:

K6 – The approaches and techniques used throughout the digital and technology solution lifecycle and their applicability to an organisation's standards and pre-existing tools.

K15 – Principles of estimating cost, and time resource constraints within digital and technology solutions activities.

K19 – Relevant legal, ethical, social and professional standards to a digital and technology solution. For example, Diversity, Accessibility, Intellectual Property, Data Protection Acts, Codes of Practice, Regulatory and Compliance frameworks.

K20 – Sustainable development approaches as applied to digital and technology solutions such as green computing.

S2 – Identify risks, determine mitigation strategies and opportunities for improvement in a digital and technology solutions project.

S4 – Initiate, design, code, test and debug a software component for a digital and technology solution.

S6 – Manage digital and technology solutions projects. For example, identifying and resolving deviations from specification, applying appropriate Project Management methodologies.

S15 – Apply relevant legal, ethical, social and professional standards to a digital and technology solution.

B1 – Has a strong work ethic and commitment in order to meet the standards required.

B2 – Reliable, objective and capable of both independent and teamworking.

B3 – Acts with integrity with respect to ethical, legal and regulatory requirements ensuring the protection of personal data, safety and security.

B4 – Commits to continuous professional development; maintaining knowledge and skills in relation to developments in digital and technology solutions that influence their work.

B5 – Interacts professionally with people from technical and non-technical backgrounds. Presents data and conclusions in an evidently truthful, concise and appropriate manner.

B7 – Maintains awareness of trends and innovations in the subject area, utilising academic literature, online sources, community interaction and conferences.

B8 – Champions diversity and inclusion in their work ensuring that digital technology solutions are accessible.

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Assessment Criteria

	Fail		Pass				
	0-29 (F)	**30-39 (E)**	**40-49 (D)**	**50-59 (C)**	**60-69 (B)**	**70-79 (A)**	**80-100 (A+)**
	Not successful	**Below required standard**	**Pass**	**Satisfactory**	**Good**	**Excellent**	**Outstanding**
Minimum pass requirements	Report must be submitted in text format (e.g., docx or pdf) and not password protected so that plagiarism checks are not circumvented.						
Introduction and background (10%)	Introduction is either missing or substantially lacking in clarity. Area of investigation not justified or wholly inappropriate.	Introduction is present but lacking clarity. Area of investigation poorly justified.	Introduction outlines project context in a basic manner but lacks some clarity. Some real-world relevance explained.	Introduction outlines project context in a mostly clear manner. Area of investigation has some real-world relevance and is adequately justified.	Introduction outlines context in a consistent and clear manner. Area of investigation has strong real-world relevance and is clearly justified.	Introduction is insightful and outlines project context in a highly consistent and clear manner. Area of investigation has strong real-world relevance and is clearly justified, **underpinned by appropriate sources**.	**In addition to A-grade:** Demonstrates a mature appreciation of a real-world issue, **well beyond what has been taught.**
Proposed Solution (20%)	Proposed solution is inappropriate or missing entirely.	Proposed solution is lacking clarity, has significant omissions, or is inappropriate.	Proposed solution is present but lacks clarity or has omissions. Appropriate visuals and/or code examples are lacking. There is an attempt to include appropriate AI technology.	Proposed solution is predominantly clear with some degree of appropriate justification. **At least one** appropriate visual or code example is included to demonstrate aspects of the solution. Proposed application of AI is appropriate and well explained.	Proposed solution is predominantly clear with appropriate justifications of design decisions. **Some appropriate** visuals and/or code examples are included to demonstrate aspects of the solution. Some evidence of an iterative approach to design documentation. Proposed application of AI is appropriate, well explained, with some justification.	Proposed solution is **very clear** with well-considered justifications of design decisions. **High-quality** visuals and/or code examples **of a professional standard** are included to demonstrate aspects of the solution. An iterative approach to the design process is clear, resulting in a refined set of design documentation. Proposed application of AI is appropriate, well explained, and fully justified.	**In addition to A-grade criteria:** Proposed solution is of outstanding quality and shows evidence of being research-informed to a standard **well beyond what has been taught.**

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	Fail		Pass				
	0-29 (F)	**30-39 (E)**	**40-49 (D)**	**50-59 (C)**	**60-69 (B)**	**70-79 (A)**	**80-100 (A+)**
Proof of Concept (30%)	Tangible outputs of the project are missing or inappropriate.	Tangible outputs of the project are present but with significant limitations.	Some tangible outputs of the project (e.g., code, configuration, user interfaces, interactive demo) are appropriately evidenced. Technical decisions may not be justified. Limited or no evidence of project management tools or approaches having been used. AI implementation attempted.	Tangible outputs of the project (e.g., code, configuration, user interfaces, interactive demo) are appropriately evidenced. Some technical decisions may be justified. Some evidence of engagement with project management tools or approaches. AI implementation achieved with some evidence of success. Appropriate use of at least **one** of the following tools: • Version control • Task tracking • Generative AI	Tangible outputs of the project (e.g., code, configuration, user interfaces, interactive demo) are appropriately evidenced. Several technical decisions are justified. Evidence of engagement with appropriate project management tools or approaches. AI implementation achieved to a good standard with clear evidence of success. Appropriate use of at least **two** of the following tools: • Version control • Task tracking • Generative AI	**In addition to B-grade criteria:** Evidence of sophisticated use of appropriate project management tools or approaches, suited to a professional context. The proof of concept (PoC) clearly demonstrates feasibility of the overall project concept. Report is supported with at least one high quality, externally linked deliverable such as a video demo, code repository, or online python notebook.	**In addition to A-grade criteria:** Demonstrates **outstanding quality** of work **well beyond what has been taught,** potentially suited for publication or funding. Innovative practices or emerging technologies were used in a sophisticated manner to support the development of the Proof of Concept.

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	Fail		Pass				
	0-29 (F)	**30-39 (E)**	**40-49 (D)**	**50-59 (C)**	**60-69 (B)**	**70-79 (A)**	**80-100 (A+)**
Privacy and Security Evaluation (20%)	Security risk assessment is missing or entirely incoherent.	Security risk assessment is attempted, but with significant omissions. Risk severity evaluation and planned mitigations are missing or incoherent.	Most aspects of the security risk assessment are completed with some appropriate risks identified, but with omissions. Risk severity evaluation and planned mitigations are missing, lacking clarity.	Most aspects of the privacy and security evaluation are completed to a sufficient standard, with some omissions. **At least two** specific examples are included. **At least two** of the following completed to a sufficient standard: • Risk severity evaluation • Planned mitigations • Evidence of at least one working mitigation	All aspects of the privacy and security evaluation are completed, with only minor omissions. **Three** specific, and highly relevant examples are included. **All** of the following completed to a sufficient standard: • Risk severity evaluation • Planned mitigations • Evidence of at least one working mitigation	**In addition to B-grade criteria:** Privacy and security evaluation draws on one or several proven analysis and/or evaluative techniques or frameworks which are referenced.	**In addition to A-grade criteria:** The security risk assessment is comprehensive and of a consistently professional standard, **with little or no suggestions for improvement**.

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Assessment Criteria

	Fail		Pass				
	0-29 (F)	**30-39 (E)**	**40-49 (D)**	**50-59 (C)**	**60-69 (B)**	**70-79 (A)**	**80-100 (A+)**
Conclusions and future plans (10%)	Conclusions are missing.	Conclusions are very limited or vague, and future plans are incoherent or missing.	Conclusions are reasonable but limited. Evaluation of the project success is attempted, with limitations.	Conclusions are mostly coherent, and success of the project is partially evaluated. **At least one** of the following is completed to an acceptable standard: • Use of feedforward to enhance final submission. • Individual learnings are evident through personal reflections. • Links to apprenticeship standard KSBs.	Conclusions are coherent and success of the project is evaluated with justifications linked back to original project aim. Some appropriate next steps are outlined. **At least two** of the following are completed to an acceptable standard: • Use of feedforward to enhance final submission. • Individual learnings are evident through personal reflections. • Links to apprenticeship standard KSBs.	Conclusions are coherent and success of the project is evaluated to an excellent standard with justifications linked back to original project aim. Next steps are clear and appropriately prioritised. **At least two** of the following are completed to a **high** standard: • **Effective** use of feedforward to enhance final submission. • Individual learnings are evident through **honest and insightful** personal reflections. • **Highly relevant** links to apprenticeship standard KSBs.	**In addition to A-grade criteria:** Future plans are realistic yet ambitious, compelling, highly detailed, and show clear intent to have a positive impact on the project area beyond the scope of the module. **All** of the following are completed to a **very high** standard: • **Effective** use of feedforward to enhance final submission. • Individual learnings are evident through **honest and insightful** personal reflections. • **Highly relevant** links to apprenticeship standard KSBs.

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Assessment Criteria

	Fail		Pass				
	0-29 (F)	**30-39 (E)**	**40-49 (D)**	**50-59 (C)**	**60-69 (B)**	**70-79 (A)**	**80-100 (A+)**
Written Communication (10%) **Evidenced by:** • Appropriate style/structure, with a high standard of design, authorship and presentation with appropriate formatting, clear concise language • Accurate spelling, grammar, and punctuation • Language appropriate to purpose and audience • Use of appropriate report writing conventions and Harvard Referencing	Work demonstrates **limited or no** ability to communicate information, ideas, and problems.	Work demonstrates an **insufficient** ability to communicate information, ideas, and problems	Work demonstrates a **sufficient** ability to: communicate information, ideas, and problems accurately and reliably utilising relevant numeracy, digital literacy, and written communication skills. Some errors or omissions. Significant parts of the writing may be overly verbose or insufficiently detailed.	Work demonstrates a **consistent and confident** ability to: communicate information, ideas, and problems accurately and reliably utilising relevant numeracy, digital literacy, and written communication skills in a manner suited to a professional context. Some errors or omissions. Some parts of the writing may be overly verbose or insufficiently detailed. Report structure may lack clarity.	Work demonstrates a **highly proficient** ability to: communicate information, ideas, and problems accurately and reliably utilising relevant numeracy, digital literacy, and written communication skills in a manner suited to a professional context. Very few errors or omissions. Appropriately detailed writing which aligns with the recommended word count. Mostly adheres to suggested report structure.	Work demonstrates a **highly accomplished** ability to: communicate information, ideas, and problems accurately and reliably utilising relevant numeracy, digital literacy, and written communication skills in a manner suited to a professional context. Limited or no errors or omissions. Clear, concise, and detailed writing which aligns with the recommended word count. Adheres closely to suggested report structure.	Work demonstrates an **exceptional** ability to: communicate information, ideas, and problems accurately and reliably utilising relevant numeracy, digital literacy, and written communication skills in a manner suited to a professional context. No noticeable errors or omissions. Clear, concise, and detailed writing which aligns with the recommended word count. Adheres closely to suggested report structure.